Name
Date
Course

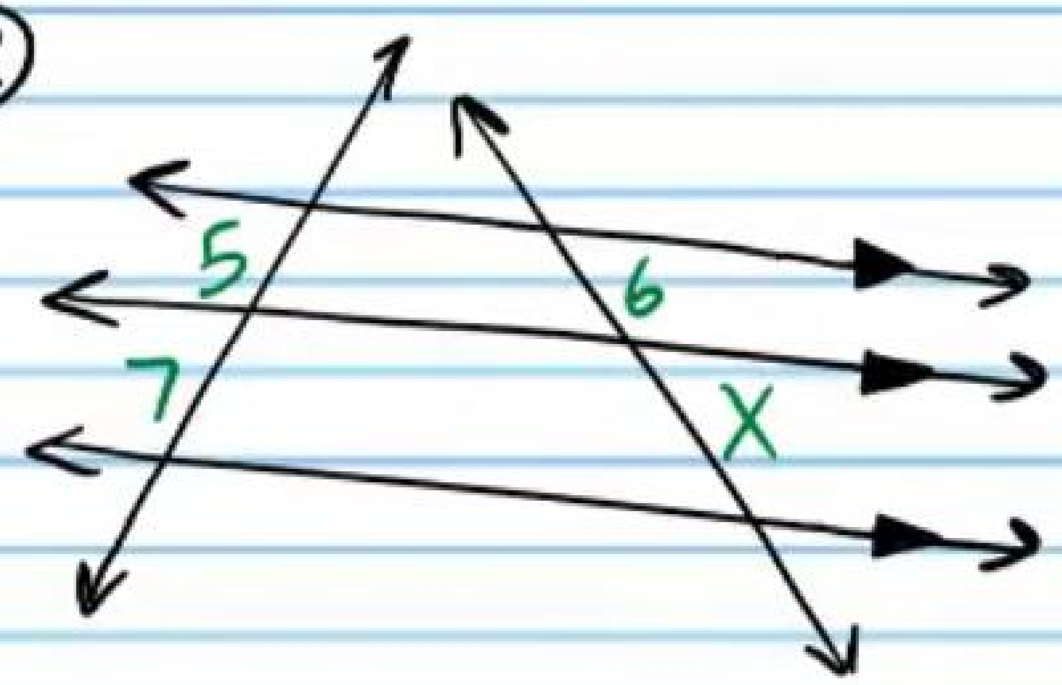
Practice Geometry Test III (Version 2)

Instructions

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show work when necessary.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) Each problem is worth four points for a total of 100 points.
V) A calculator may be useful for some problems on this test.

①

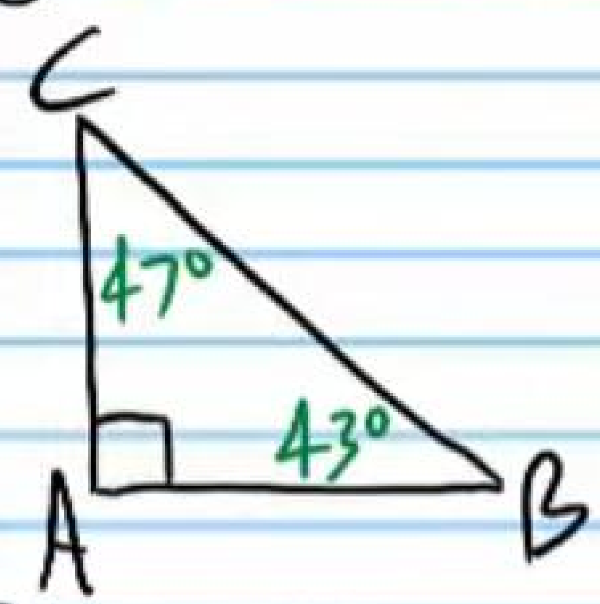
②



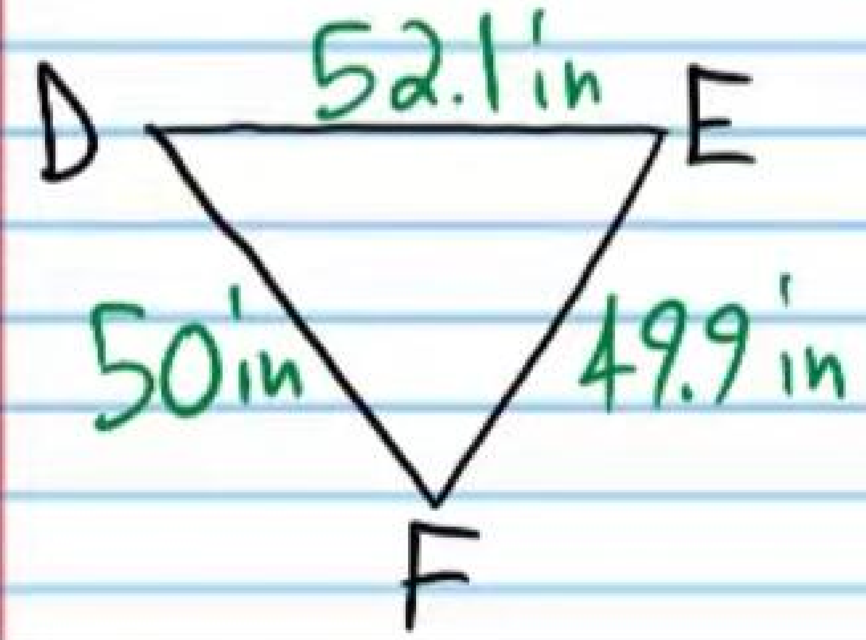
③ If the pair of numbers below represents side lengths of a triangle, write an inequality that shows the possible lengths of the third side x.

4, 10

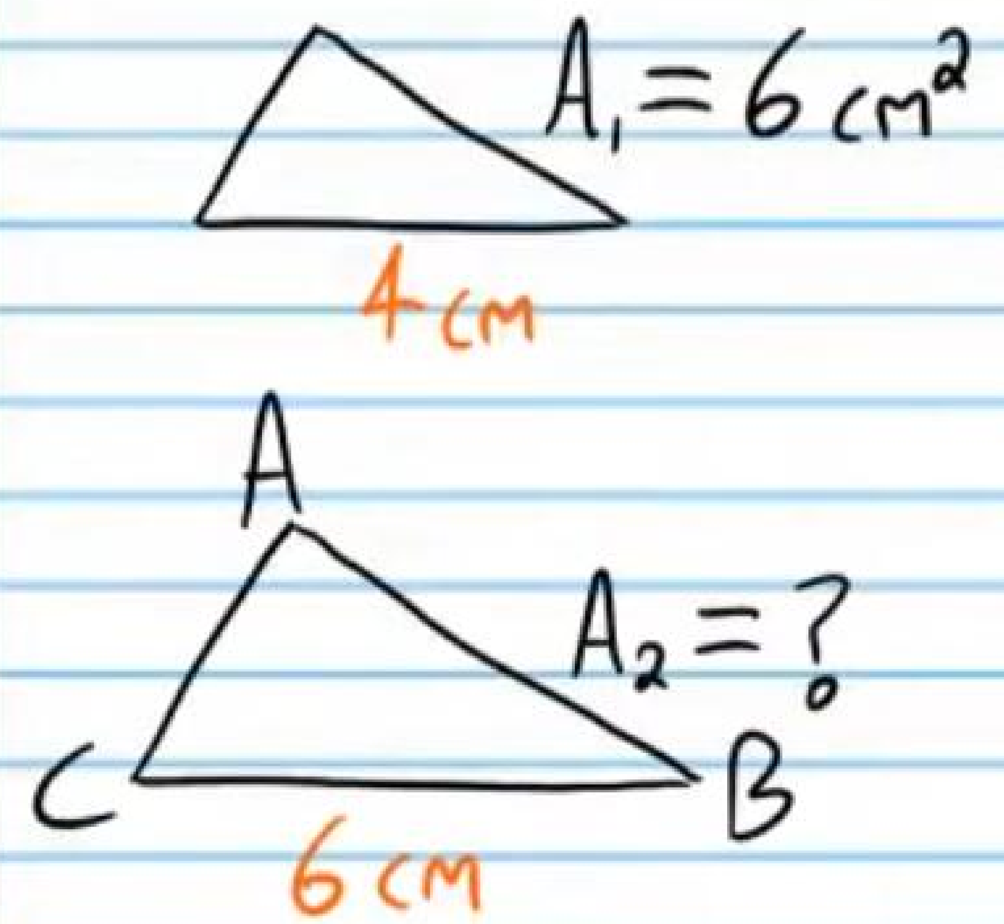
④ List the sides of the triangle from shortest to longest.



⑤ List the angles of the triangle from least to greatest.



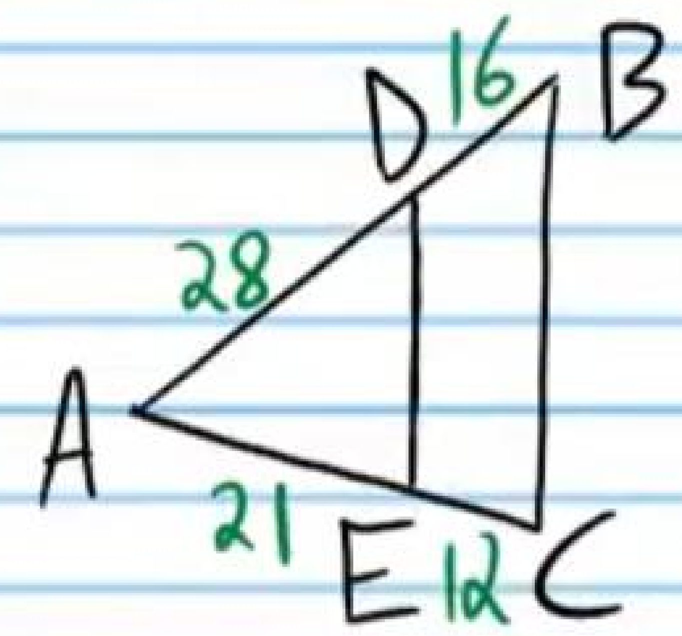
⑥ Find the area of $\triangle ABC$ if the two polygons below are similar. Show your proportion.



⑦ Is it possible to make a triangle with the group of side lengths below? If not, write the inequality that shows that it is not possible.

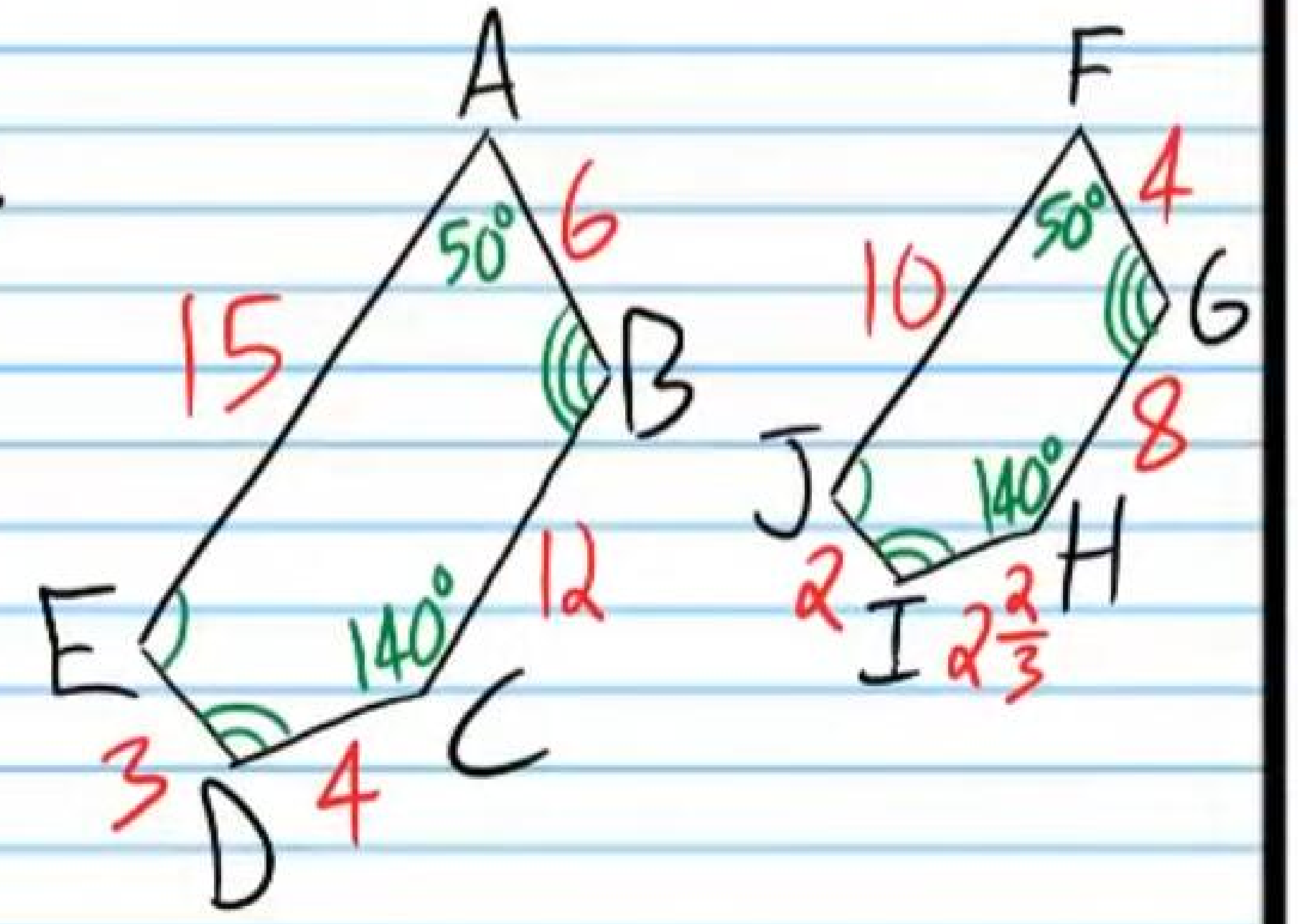
5, 3, 9

⑧ Is $\overline{DE} \parallel \overline{BC}$? Show the proportions that prove your answer.



Use the polygons below to answer problem #'s 9-12.

9) Show that the polygons are similar.



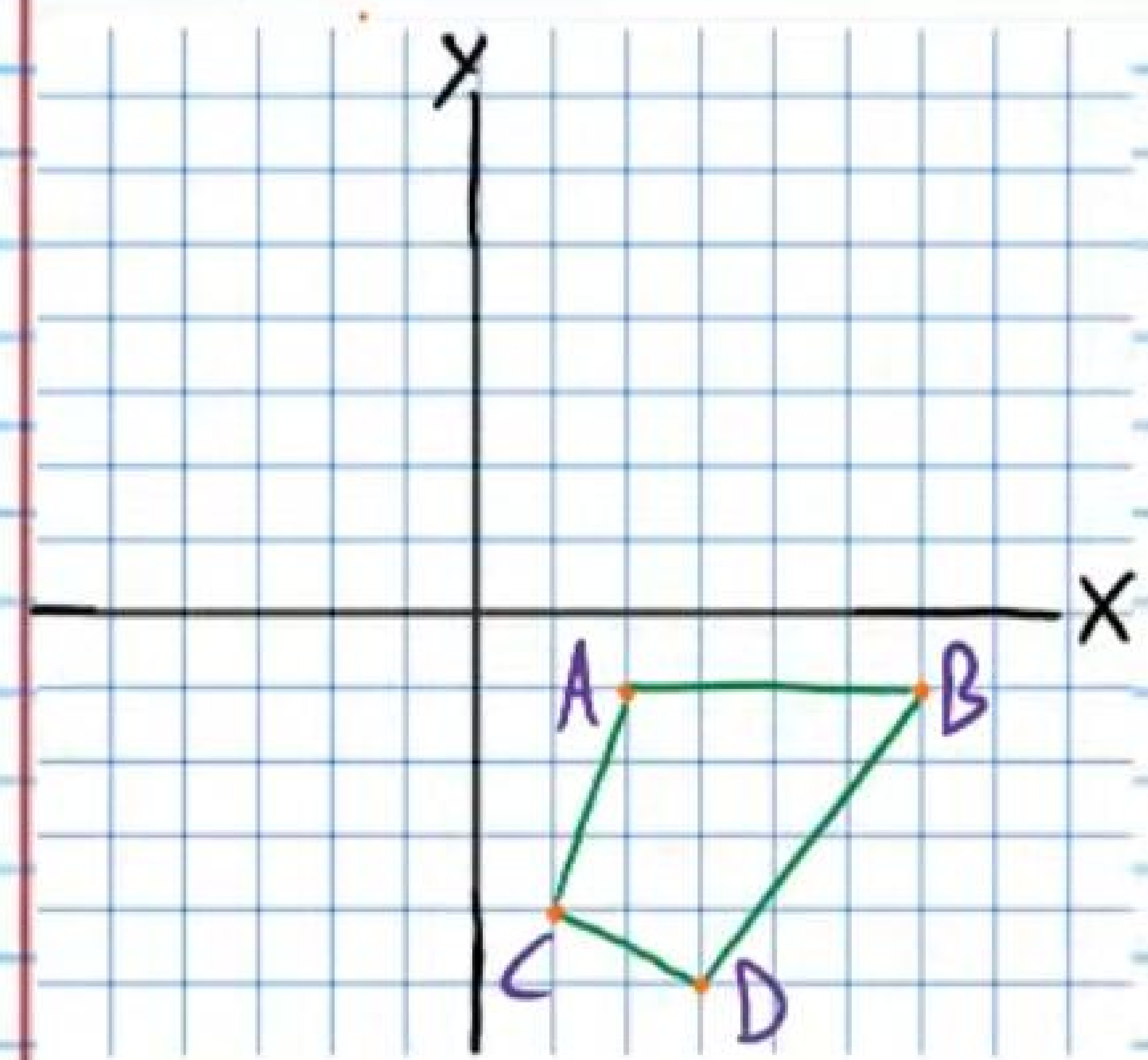
10) Write a similarity statement:
Write in alphabetical order.

11) Write the scale factor:

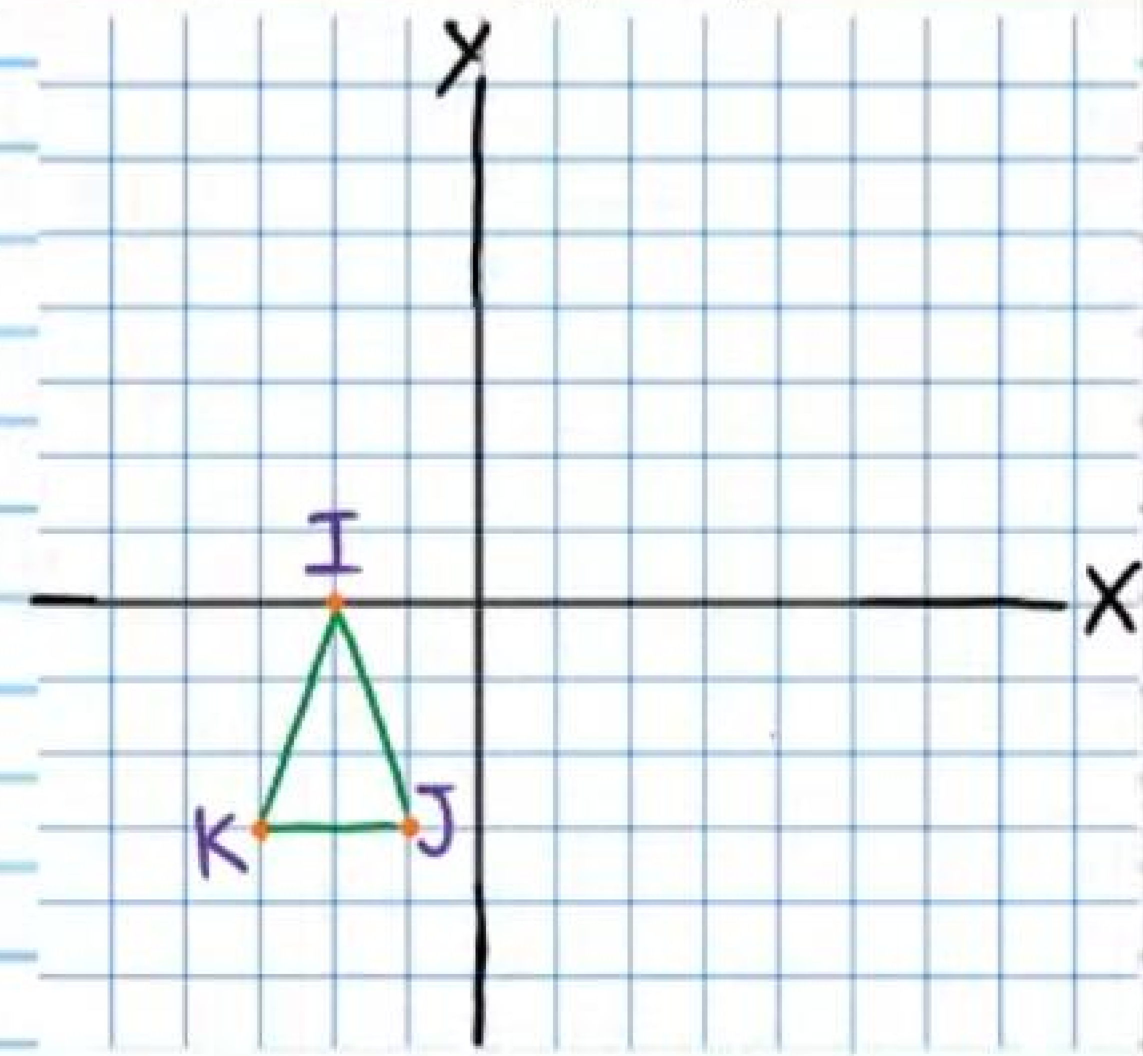
12) Write the scale factor in a different way
(Hint: Remember, there are two ways to write a scale factor).

Given the "pre-image," graph the "image" using the given transformation.

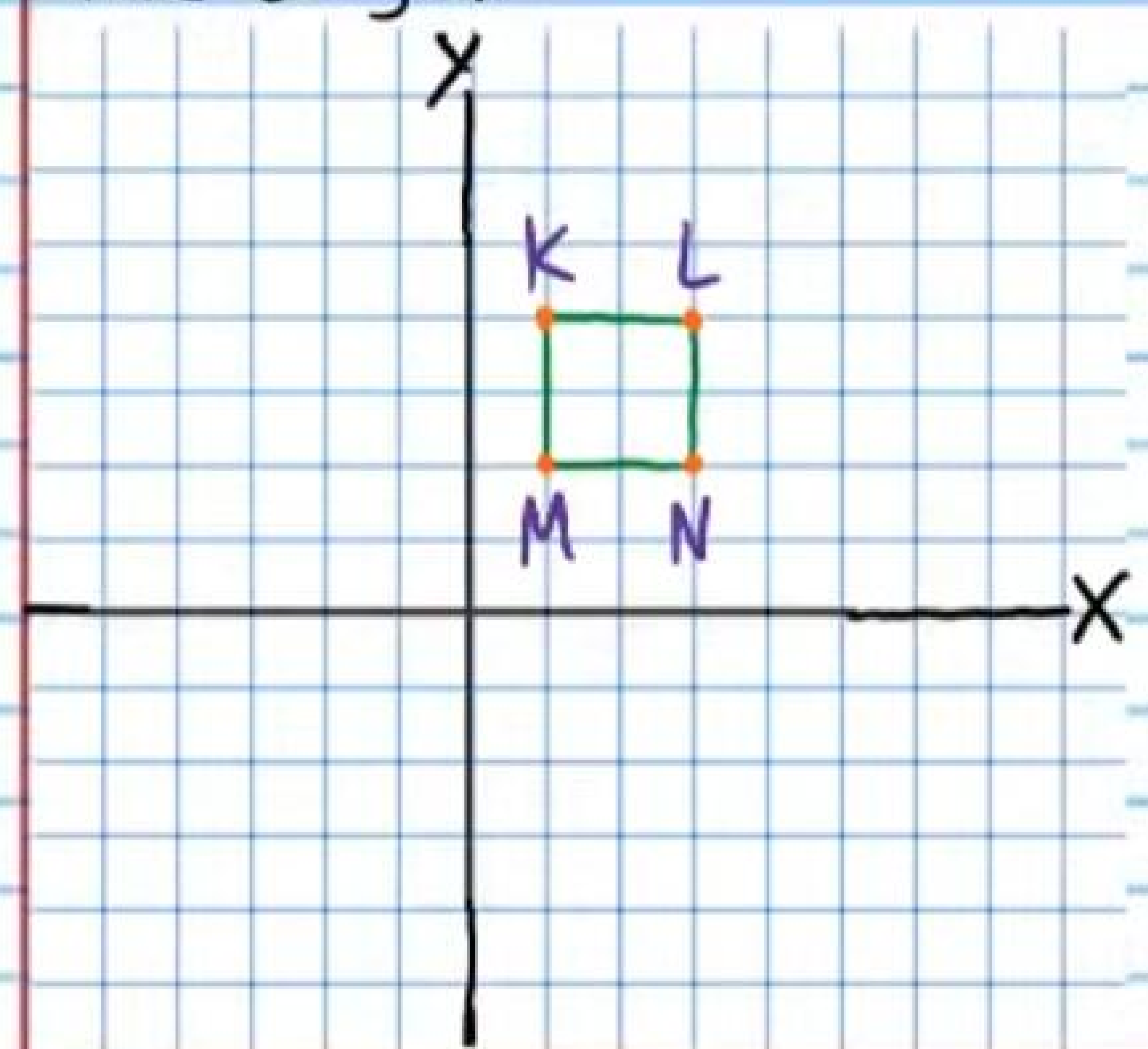
13) Reflection about the x-axis



14) Translation: $(x,y) \rightarrow (x-2, y+1)$

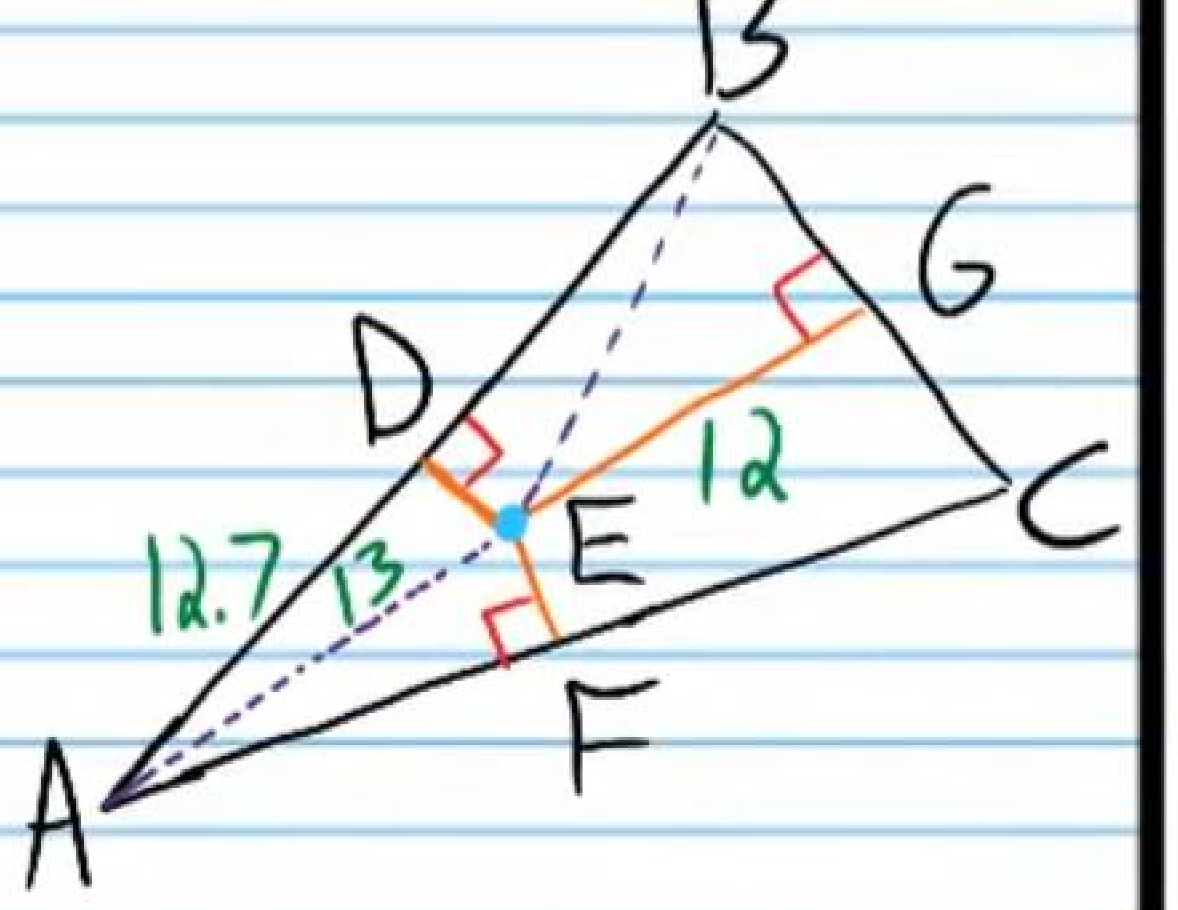


15) Rotation: $-270°$ about the origin.



16. If point E is the circumcenter of the triangle below, find the quantities listed.

i) DB



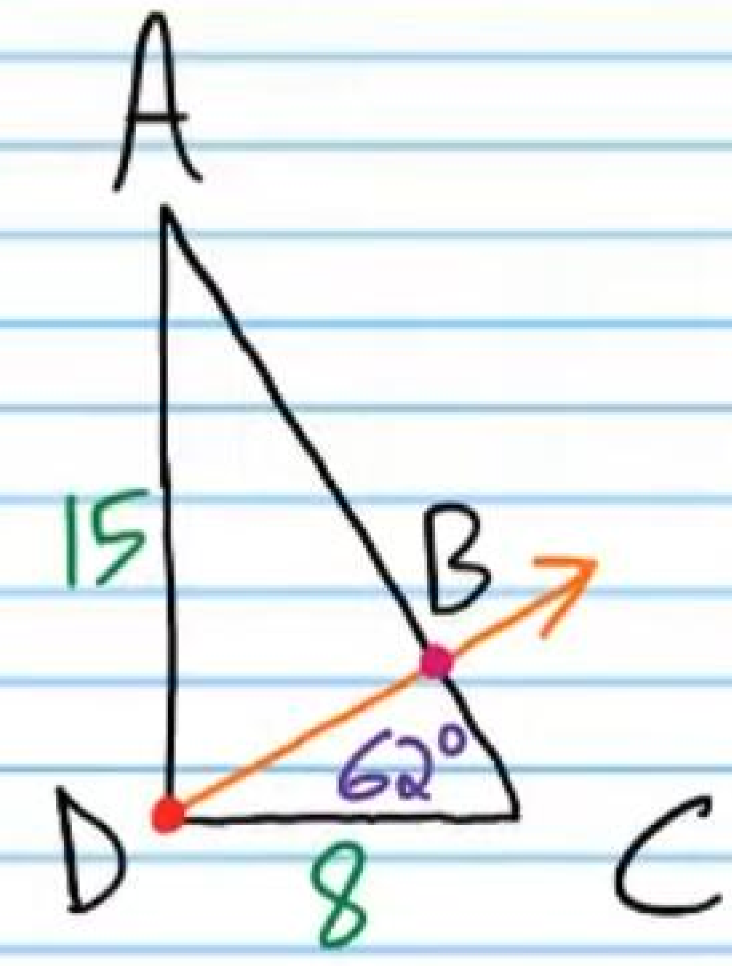
ii) EB

iii) BG

iv) BC

17. If point D is the orthocenter of the triangle below, find the quantities listed.

i) $m\angle BDC$



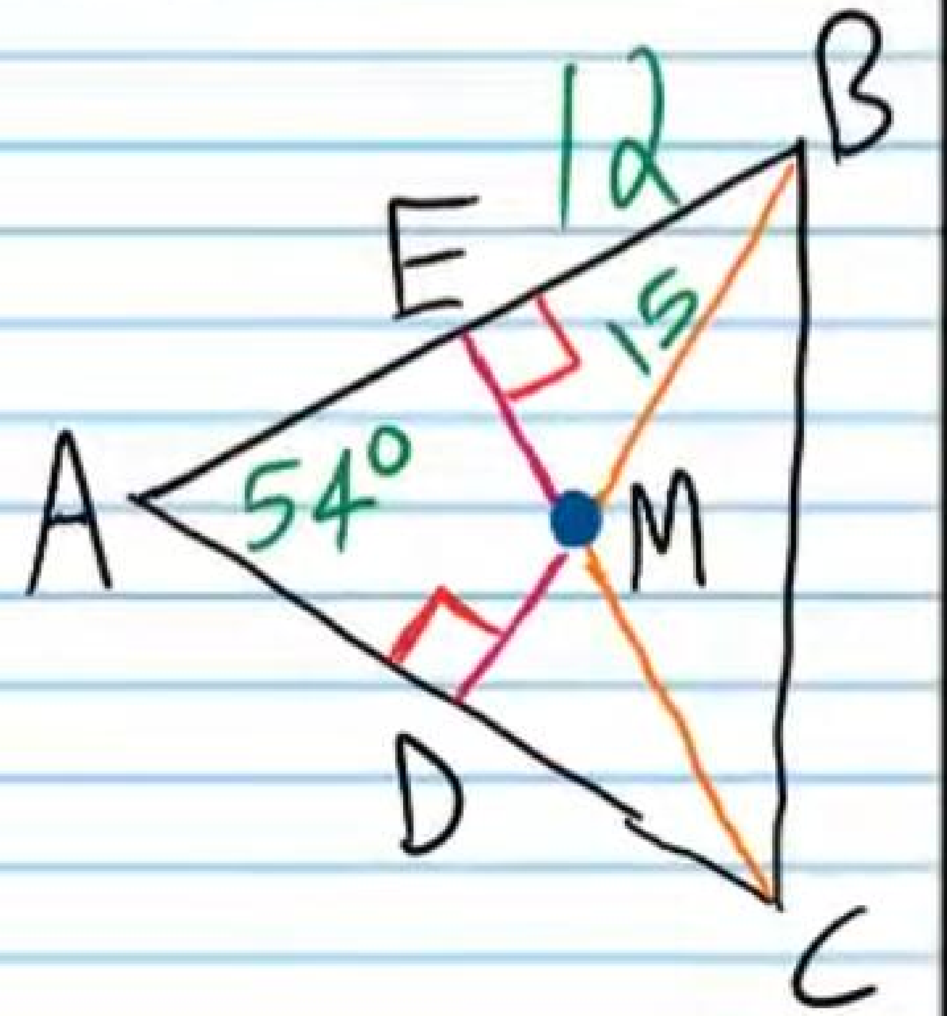
ii) $m\angle DBA$

iii) AC

iv) $m\angle ADC$

(18) If point M is the incenter of the triangle below, find the quantities listed.

i) If $m\angle DCB = 52°$, find $m\angle EBM$.



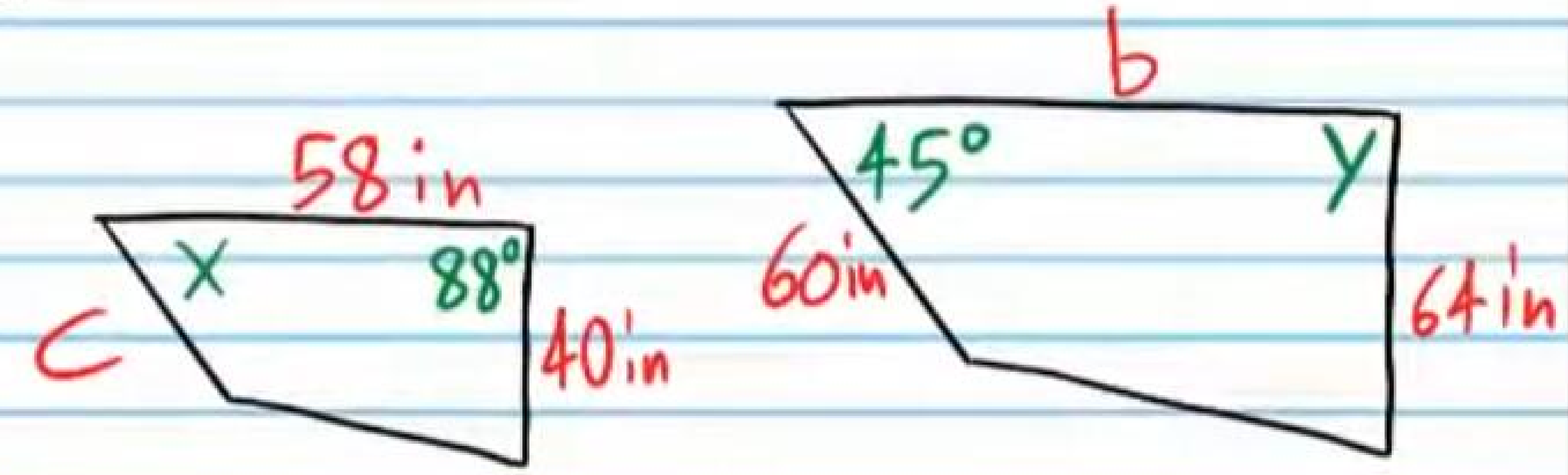
ii) $m\angle EBC$.

iii) EM

iv) DM

(19) The polygons below are similar. Find the unknowns.

i) Find X.

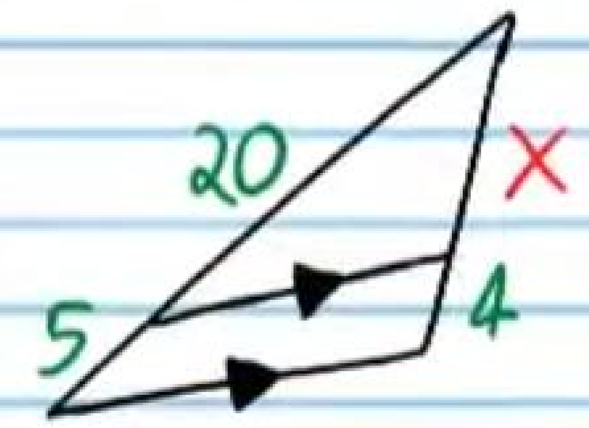


ii) Find y.

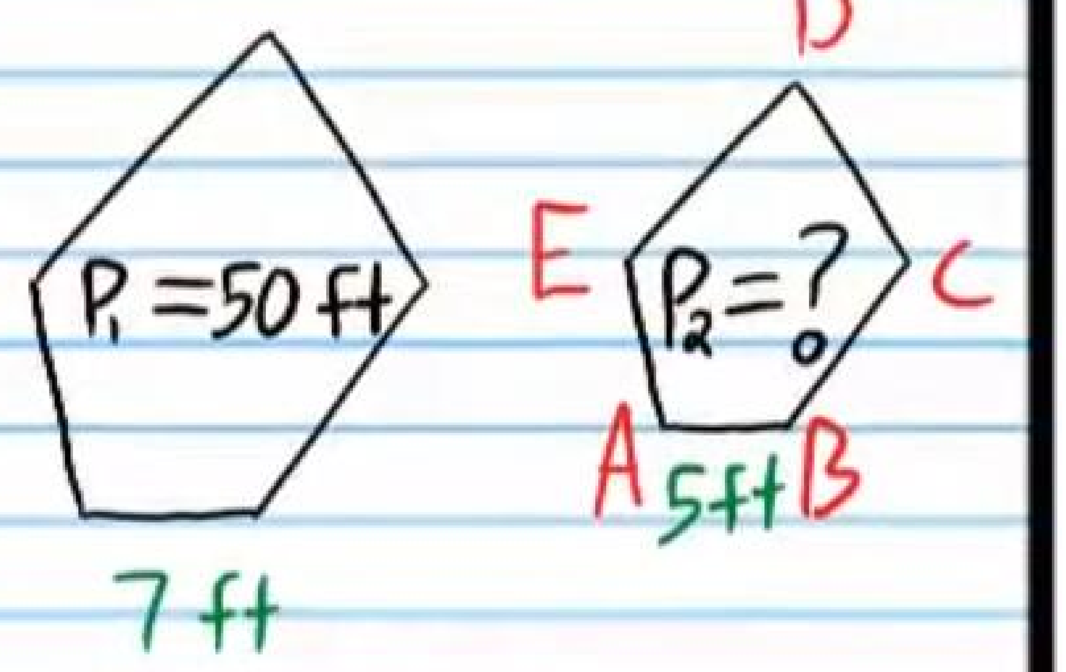
iii) C

iv) b

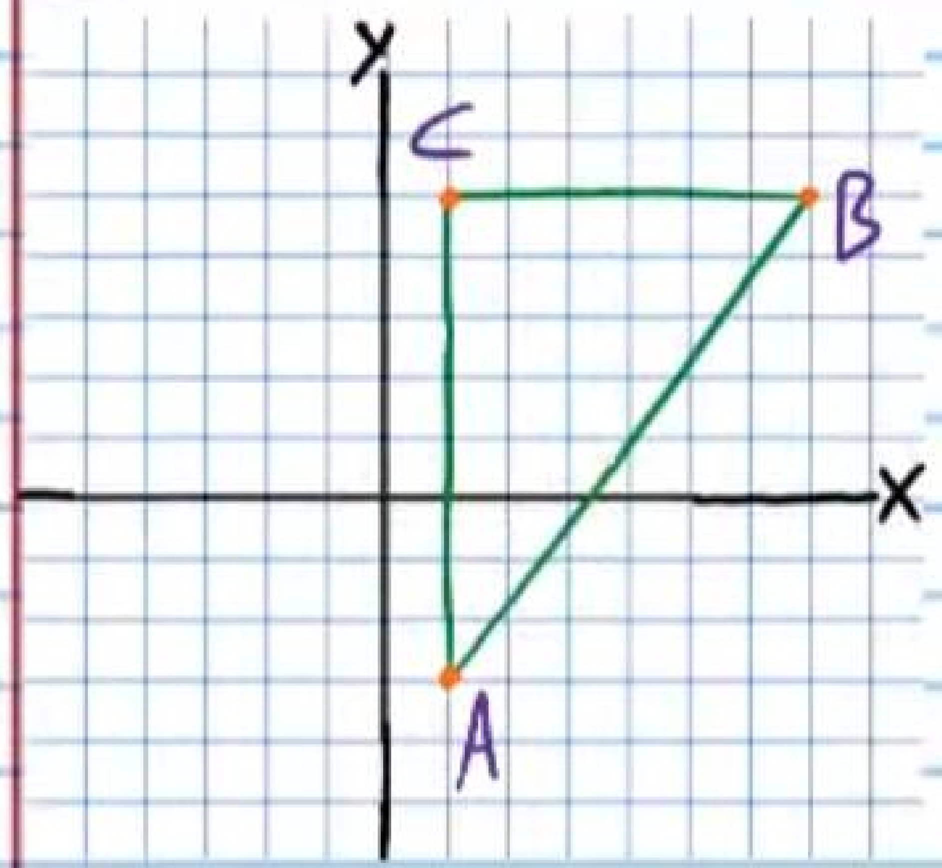
(20) Find X.



(21) Find the perimeter of ABCDE if the polygons below are similar.

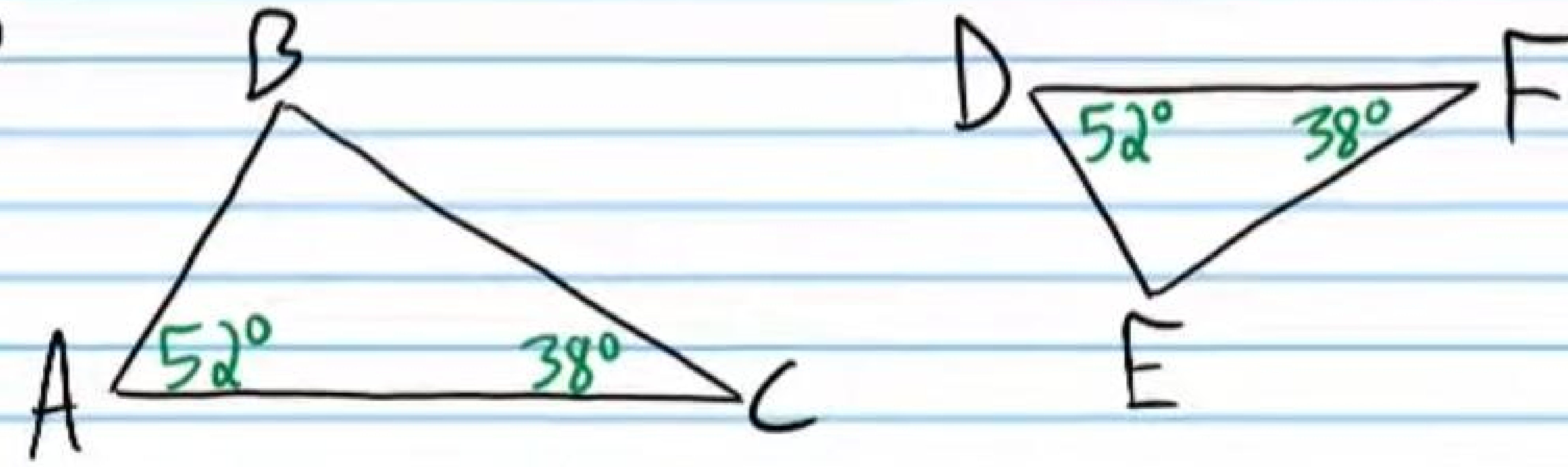


(22) Find the coordinates of the circumcenter of the triangle below.



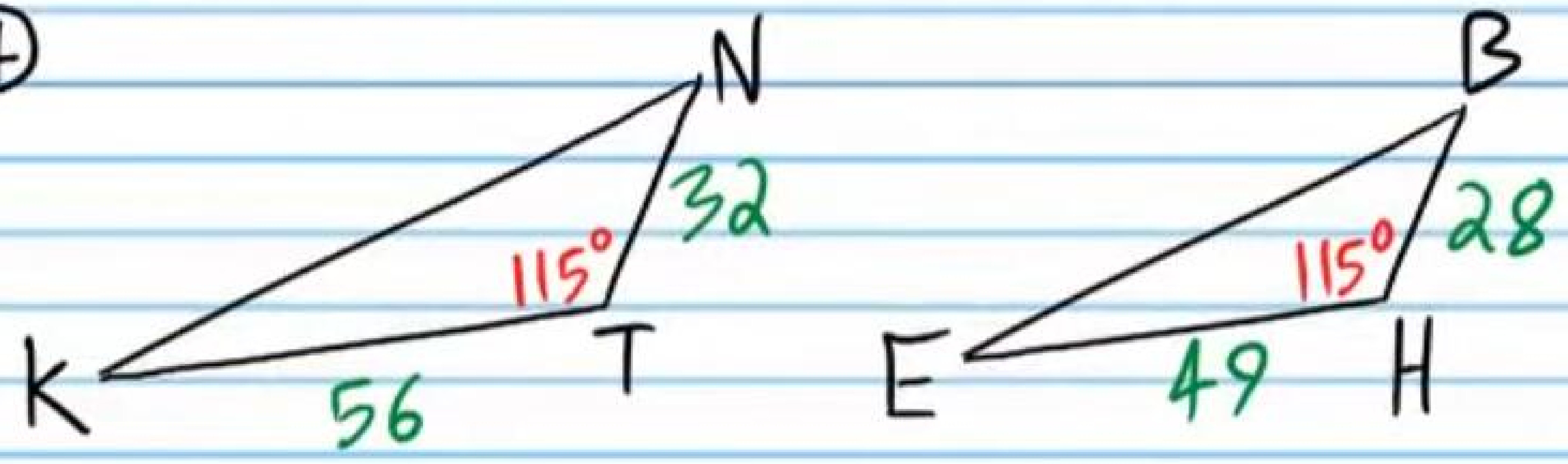
In the following problems, write two-column proofs showing that the triangles are similar. You might not use the same number of steps written in Roman numerals. You may use more or less.

(23)



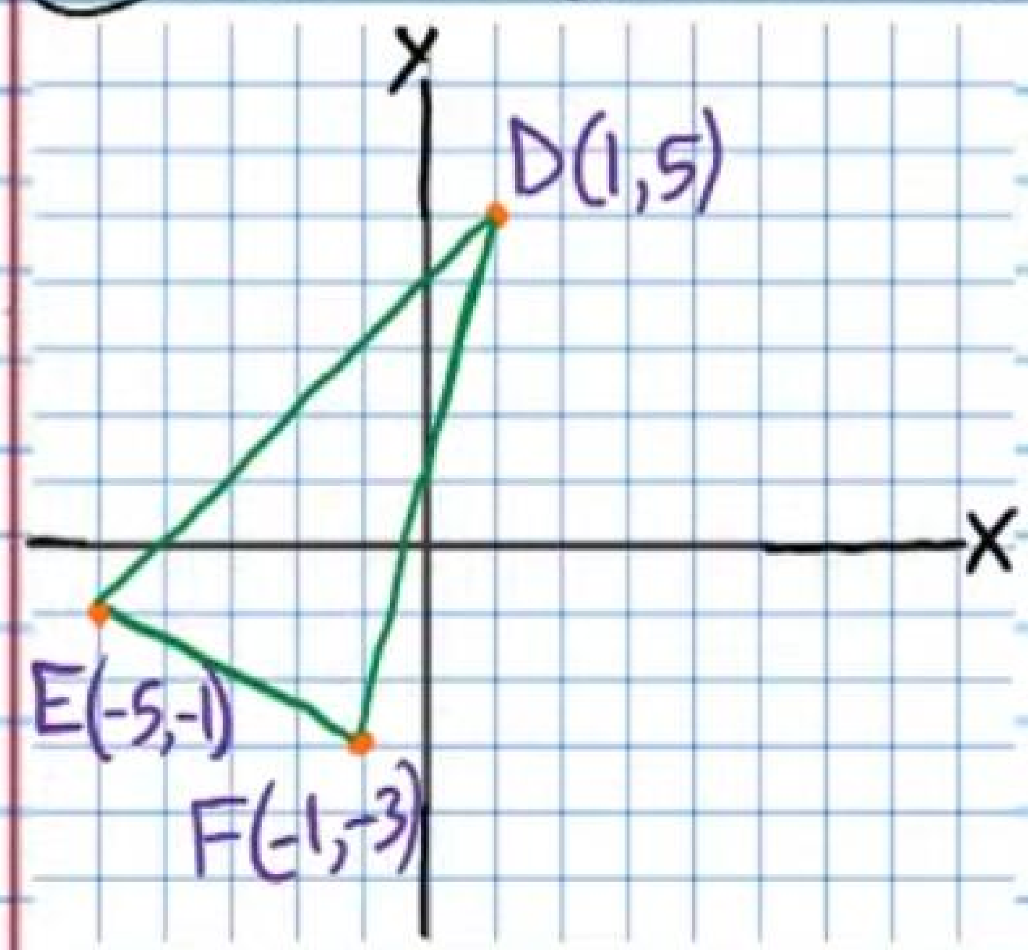
Statement	Reason
I)	
II)	
III)	
IV)	

(24)



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

(25) Find the coordinates of the centroid of the triangle below.



Practice Geometry Test III (Version 2) Answers

1. $\boxed{y = 14}$

2. $\boxed{x = 8\frac{2}{5} \text{ or } 8.4}$

3. $\boxed{6 < x < 14}$

4. $\boxed{CA,\ AB,\ CB}$

5. $\boxed{\angle D,\ \angle E,\ \angle F}$

6. $\boxed{13.5 \text{ cm}^2}$

7. $\boxed{\text{No}}$ $\boxed{\begin{array}{c} 3+5 \ngtr 9 \\ 8 \ngtr 9 \end{array}}$

8. $\boxed{\text{Yes}}$ $\boxed{\begin{array}{ccc} \frac{28}{16} = \frac{21}{12} & \text{or} & \frac{16}{28} = \frac{12}{21} \\ \frac{7}{4} = \frac{7}{4} & & \frac{4}{7} = \frac{4}{7} \end{array}}$

9. I) $\boxed{\angle A \cong \angle F,\ \angle E \cong \angle J,\ \angle D \cong \angle I,\ \angle C \cong \angle H,\ \& \ \angle B \cong \angle G}$

II) $\frac{15}{10} = \frac{6}{4} = \frac{3}{2} = \frac{12}{8} = \frac{4}{2\frac{2}{3}}$

$\frac{3}{2} = \frac{3}{2} = \frac{3}{2} = \frac{3}{2} = \frac{3}{2}$

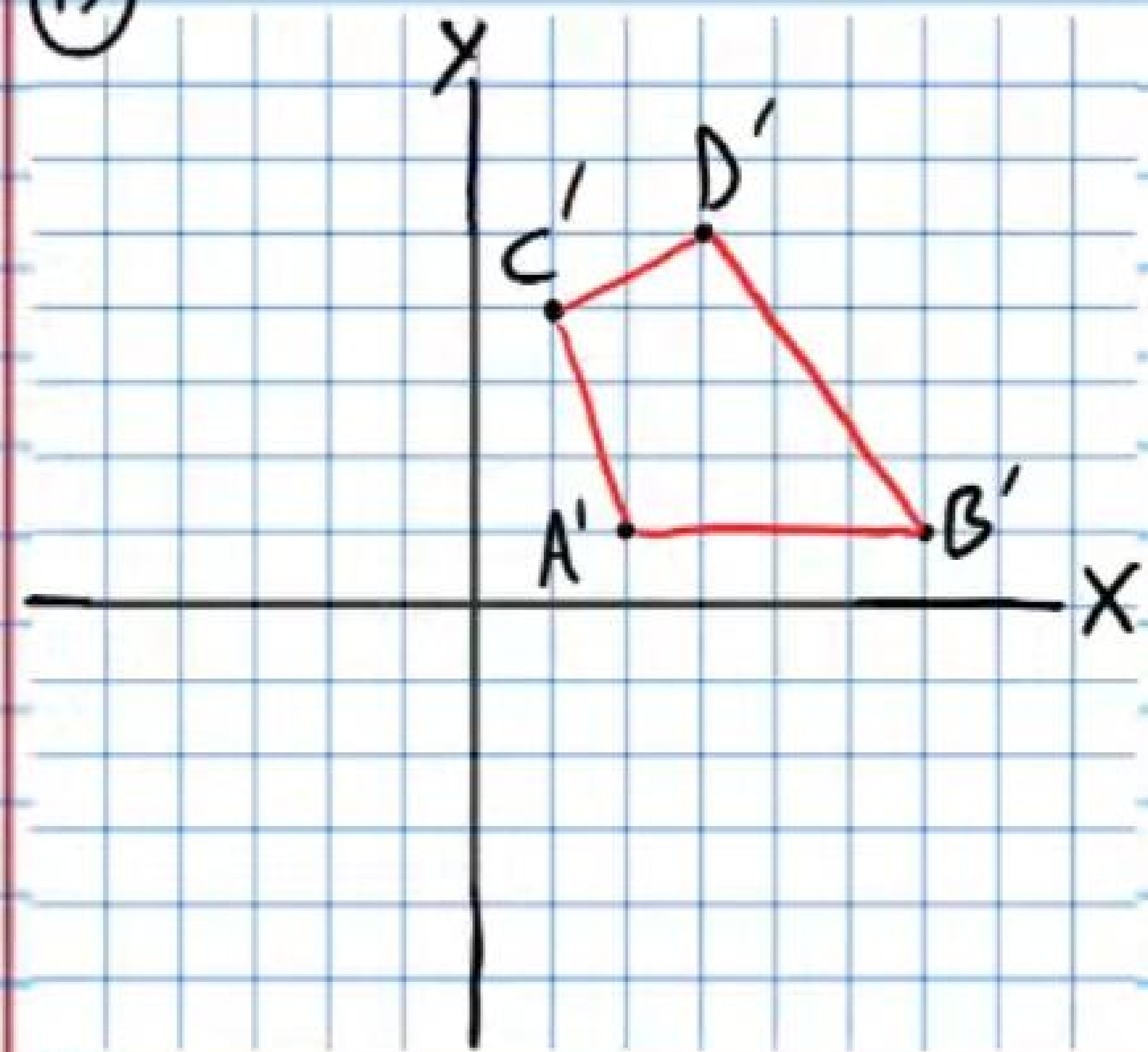
$$\boxed{\frac{AE}{FJ} = \frac{AB}{FG} = \frac{ED}{JI} = \frac{BC}{GH} = \frac{DC}{IH}}$$

10. $\boxed{ABCDE \sim FGHIJ}$

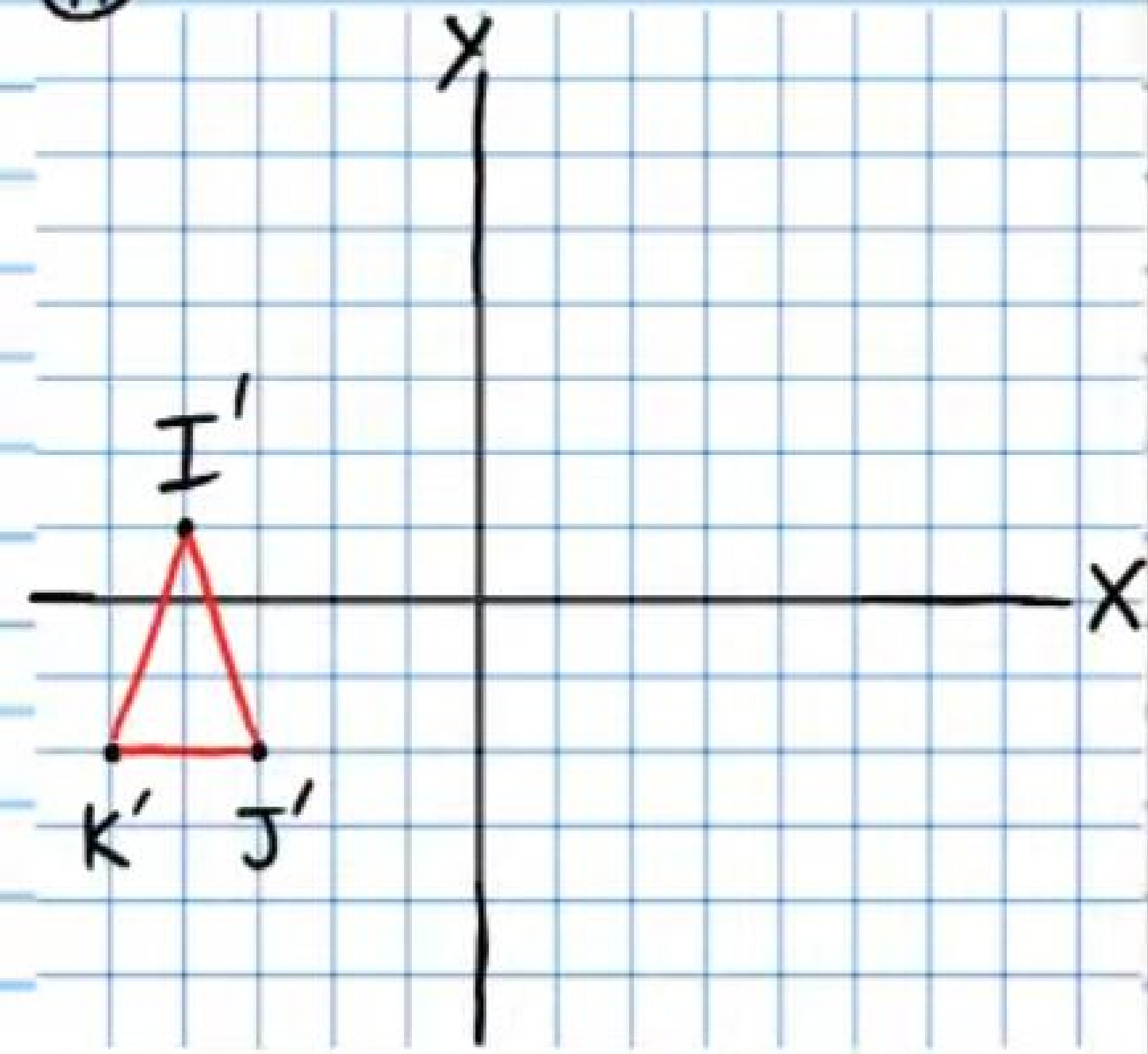
11. $\boxed{\frac{2}{3} \text{ or } \frac{3}{2}}$

12. If you wrote $\frac{3}{2}$ for problem #11, then $\frac{2}{3}$ is the answer for this problem. If you wrote $2/3$ for problem #11, then $3/2$ is the answer for this problem.

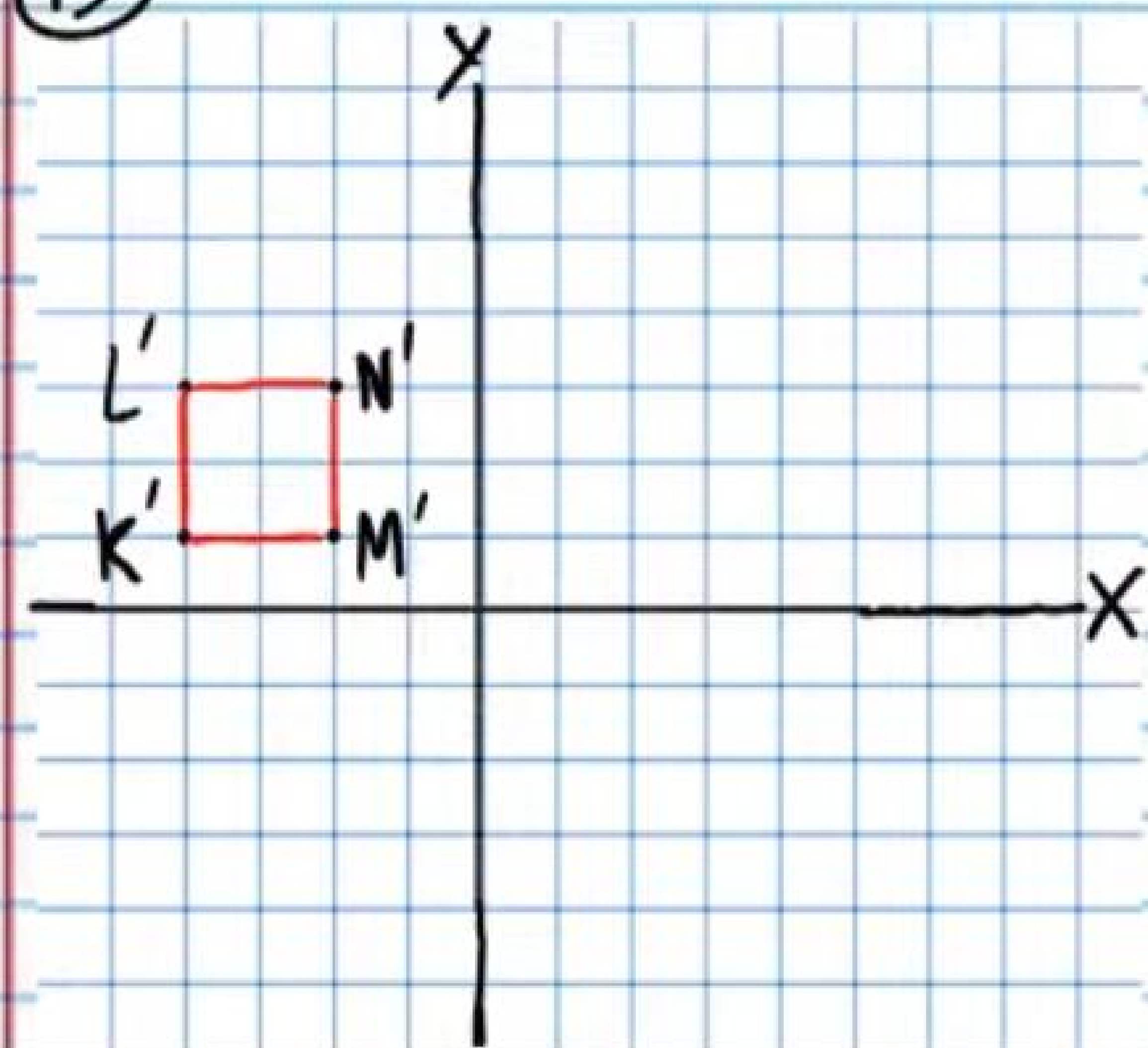
13.



14.



15.



⑯ i) $\boxed{12.7}$ ii) $\boxed{13}$ iii) $\boxed{5}$ iv) $\boxed{10}$

⑰ i) $\boxed{28^\circ}$ ii) $\boxed{90^\circ}$ iii) $\boxed{17}$ iv) $\boxed{90^\circ}$

⑱ i) $\boxed{37^\circ}$ ii) $\boxed{74^\circ}$ iii) $\boxed{9}$ iv) $\boxed{9}$

⑲ i) $\boxed{x=45^\circ}$ ii) $\boxed{y=88^\circ}$

iii) $\boxed{c = 75/2 \text{ in or } 37.5 \text{ in}}$ iv) $\boxed{b = 464/5 \text{ in or } 92.8 \text{ in}}$

⑳ $\boxed{x=16}$

㉑ $\boxed{P_2 = \frac{250}{7} \text{ ft or } 35\frac{5}{7} \text{ ft}}$

㉒ $\boxed{(4,1)}$

㉓

Statement	Reason
I) $m\angle A = 52^\circ$, $m\angle C = 38^\circ$, $m\angle D = 52^\circ$, & $m\angle F = 38^\circ$.	Given
II) $m\angle A = m\angle D$ & $m\angle C = m\angle F$	Trans. Prop of =.
III) $\angle A \cong \angle D$ & $\angle C \cong \angle F$	Def. of $\cong$ $\angle$'s.
IV) $\Delta ABC \sim \Delta DEF$	AA

㉔

Statement	Reason
I) $NT = 32$, $KT = 56$, $m\angle T = 115^\circ$, $BH = 28$, $EH = 49$, & $m\angle H = 115^\circ$	Given
II) $m\angle T = m\angle H$	Trans. Prop of =.
III) $\angle T \cong \angle H$	Def. of $\cong$ $\angle$'s.
IV) $\frac{KT}{EH} = \frac{56}{49}$ & $\frac{NT}{BH} = \frac{32}{28}$.	Subs. Prop of =.
V) $\frac{KT}{EH} = \frac{8}{7}$ & $\frac{NT}{BH} = \frac{8}{7}$.	Simplification.
VI) $\frac{KT}{EH} = \frac{NT}{BH}$.	Trans. Prop of =.
VII) $\Delta KTN \cong \Delta EHB$	SAS

㉕ $\boxed{\left(-\frac{5}{3}, \frac{1}{3}\right)}$ or $\boxed{\left(-1\frac{2}{3}, \frac{1}{3}\right)}$